

Final Paper Tables

All comparisons within each table use the **same source model** (fair setting), except where a “reference” row explicitly notes a different source. $\pm$ std = 3-seed mean $\pm$ sample stdev (seeds 0, 42, 123).

Method naming

Name	What it is
V2 (STR)	Our component: shared-trunk stochastic restore (backbone/FPN restore prob $\times$ 0.1). Plug-in, works with any restore-based CTTA (CoTTA, AMROD, ours).
CTPV	Our component: cross-task pseudo-label verification. Seg head vetoes teacher detections whose interior does not classify as the corresponding seg class.
CTCMT_Det	Our full single-task detection method (soft-CE + dyn-threshold pseudo-labels + score-EM gate + V2). Sits on Mask R-CNN source.
CTCMT_Seg	Our full single-task segmentation method (soft-CE from EMA teacher + V2 + per-class prototype anchor + small-scale aug).
CTCMT-MTL + V2 + CTPV	Our full multi-task method: shared backbone + det/seg heads with soft-CE teacher, cross-task contrastive (CT-CL), cross-task consistency (CT-CR), V2 restore, and CTPV pseudo-label verification.
AMROD + V2	The AMROD framework (Wei et al. ESWA 2026) with our V2 plug-in. Demonstrates V2 is a general-purpose improvement, not tied to our MTL system.

Table 1 — Single-task Object Detection

Source: Mask R-CNN R-50 FPN (Cityscapes 8-class det, AP50 on Cityscapes val = 41.2).

ACDC continual (fog $\rightarrow$ night $\rightarrow$ rain $\rightarrow$ snow)

Method	Fog	Night	Rain	Snow	Mean AP50
Source only	50.20	13.45	30.70	37.13	32.87
W3TTA-OD (Yoo et al., CVPR 2024)	58.38	18.33	29.81	30.29	34.20

Method	Fog	Night	Rain	Snow	Mean AP50
AMROD (Wei et al., ESWA 2026)	51.38	15.23	33.74	40.99	35.34
AMROD + V2 (V2 = ours)	52.16	15.98	34.30	41.78	36.06
CTCMT_Det (ours, n=3)	52.04 $\pm$ 0.37	15.27 $\pm$ 0.20	34.43 $\pm$ 0.21	43.31 $\pm$ 0.41	36.26 $\pm$ 0.14

CTCMT_Det is the native single-task detection variant of CTCMT-MTL: same soft-CE + dyn-threshold pseudo-labels + V2 restore, no seg branch. It beats AMROD by +0.92 AP50 and edges past AMROD+V2 by +0.20 AP50 on the mean. The largest win is on **snow** (+1.53 vs AMROD+V2, +2.32 vs AMROD), showing the expected anti-forgetting behaviour on the tail of the continual sequence. Multi-seed variance is tight ($\pm$ 0.14 on the mean).

The AMROD+V2 row is retained to demonstrate that our V2 component is a **general-purpose plug-in** — it improves the published SOTA method by +0.72 AP50 with a one-line config change.

Foggy Cityscapes (single-shot)

Method	AP50
Source only	34.04
AMROD (ESWA 2026)	35.25
AMROD + V2 (V2 = ours)	35.58

Table 2 — Single-task Semantic Segmentation

Source: Semantic FPN R-50 (Cityscapes 19-class, mIoU on Cityscapes val = 71.75).

ACDC continual (fog $\rightarrow$ night $\rightarrow$ rain $\rightarrow$ snow)

Method	Fog	Night	Rain	Snow	Mean mIoU
Source only	45.82	22.60	41.88	36.82	36.78
TENT (Wang et al., ICLR 2021)	3.54	1.56	4.13	4.56	3.45
CoTTA (Wang et al., CVPR 2022)	46.24	22.24	41.27	35.79	36.39

Method	Fog	Night	Rain	Snow	Mean mIoU
CTCMT_Seg (ours)	46.38	22.81	42.68	38.82	37.67

CTCMT_Seg beats CoTTA on all 4 weathers (+1.28 mIoU mean). TENT collapses in the continual setting (well-known failure without episodic reset).

Table 3 — Multi-Task Learning (Detection + Segmentation)

Source: Panoptic-FPN R-50 MTL (Cityscapes; det AP50 = 41.5, seg mIoU = 69.24).

The MTL source is intentionally used to prove that **one shared model matches or beats specialised single-task models** — the core claim of MTL-CTTA.

ACDC continual — Detection (bbox AP50)

Method	Source	Fog	Night	Rain	Snow	Mean
Source only (PFN MTL)	PFN	50.63	15.19	30.59	35.46	32.97
CTCMT-det (single-task, PFN)	PFN	51.09	15.77	32.41	37.94	34.30
<i>CTCMT_DeMR- (single-task, MR-CNN, n=3)</i>	<i>CNN</i>	<i>52.04 ± 0.37</i>	<i>15.27 ± 0.20</i>	<i>34.43 ± 0.21</i>	<i>43.31 ± 0.41</i>	<i>36.26 ± 0.14</i>
CTCMT-MTL + V2 + CTPV (ours, n=3)	PFN	52.86 ± 0.02	17.00 ± 0.06	35.81 ± 0.41	44.05 ± 0.33	37.43 ± 0.07

Our MTL beats the same-source single-task detector by **+3.13 AP50**, and even beats the specialised single-task SOTA using a stronger source (Mask R-CNN) by **+1.17 AP50**.

ACDC continual — Segmentation (mIoU)

Method	Source	Fog	Night	Rain	Snow	Mean
Source only (PFN MTL)	PFN	38.46	18.78	32.39	32.14	30.44
CTCMT-seg (single-task, PFN)	PFN	38.80	19.10	33.53	34.33	31.44
<i>CTCMT_Seg^{5F}</i> (single-task, Sem-FPN)		<i>46.38</i>	<i>22.81</i>	<i>42.68</i>	<i>38.82</i>	<i>37.67</i>
CTCMT-MTL + V2 + CTPV (ours, n=3)	PFN	39.89 ± 0.04	20.17 ± 0.05	37.45 ± 0.12	40.16 ± 0.13	34.42 ± 0.06

Our MTL beats the same-source single-task segmenter by **+2.98 mIoU**. The absolute gap vs CTCMT_Seg on Semantic FPN (−3.25 mIoU) is explained by the **source-model gap**: PFN MTL source is 2.5 mIoU below Sem-FPN on Cityscapes val (69.24 vs 71.75).

Foggy Cityscapes (single-shot)

Method	AP50	mIoU
Source only (PFN MTL)	32.15	44.12
CTCMT-MTL + V2 + CTPV (ours)	33.19	49.97

Reading the tables

- **Bold rows** are ours (or contain our contribution).
- **Italic rows** in Table 3 are reference numbers using a different (single-task-optimised) source model — they can’t be direct competitors to MTL but they anchor the reader on what specialised methods can achieve.
- All ACDC numbers are the **continual** setting (adapt fog → night → rain → snow, no reset between domains). Foggy Cityscapes is single-shot.
- Multi-seed variance ($\pm$ std, n=3) is reported for Table 3 because the MTL system is the paper’s primary contribution.